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Inventor(s)

Fujimoto; Kei et al.

CONNECTOR UNIT

Abstract

A connector unit includes: a terminal fitting; a housing configured to accommodate the terminal fitting; a fixing fitting embedded in the housing and configured to fasten and fix the terminal fitting and an external conductive component; a metal frame member on which the housing is mounted; and a heat transfer member configured to thermally connect the frame member and a part of the fixing fitting exposed from the housing. The heat transfer member is sandwiched between the part of the fixing fitting and the frame member, and is flexible enough to deform to fit a shape of a gap between the part of the fixing fitting and the frame member.

Inventors: Fujimoto; Kei (Makinohara-shi, JP), Muro; Takashi (Makinohara-shi, JP)

Applicant: YAZAKI CORPORATION (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is based upon and claims the benefit of priority from prior Japanese patent application No. 2024-036052 filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field of the Invention

[0002] The present disclosure relates to a connector unit including a terminal fitting, a housing that houses the terminal fitting, a fixing fitting that is embedded in the housing and that fastens and fixes the terminal fitting and an external conductive component, a metal frame member to which the housing is attached, and a heat transfer member that thermally connects the frame member and a part of the fixing fitting exposed from the housing.

2. Description of the Related Art

[0003] In the related art, various connectors have been proposed for use in a power cable for supplying power from a power source mounted on a vehicle or the like to an electrical load (for example, see JP2022-83460A).

[0004] In the above-described types of connectors, a terminal fitting and conductive components such as an electric wire and a bus bar are generally electrically connected in a housing. Although the connection portion between the terminal fitting and these conductive components is a portion where a large amount of Joule heat is generated at the time of energization due to the large contact resistance, the connection portion is isolated from the outside by being covered with the housing. Therefore, it is difficult to dissipate heat from the connection portion to the outside. Due to such difficulties in dissipating heat, the temperature of the connector may rise excessively at the time of energization, which may cause deterioration of the components that constitute the connector. On the other hand, simply attaching a dedicated heat dissipation member (for example, a heat dissipation fin) to the outside of the connector is undesirable because the heat dissipation member may hinder the miniaturization of the connector and because the space for providing the connector inside the vehicle body is limited.

SUMMARY

[0005] An object of the present disclosure is to provide a connector unit capable of improving the heat dissipation property while avoiding an increase in size of the connector unit.

[0006] In order to achieve the above object, a connector unit according to the present disclosure has the following features.

[0007] According to an aspect of the present disclosure, there is provided a connector unit including: a terminal fitting; a housing configured to accommodate the terminal fitting; a fixing fitting embedded in the housing and configured to fasten and fix the terminal fitting and an external conductive component; a metal frame member on which the housing is mounted; and a heat transfer member configured to thermally connect the frame member and a part of the fixing fitting exposed from the housing, in which the heat transfer member is sandwiched between the part of the fixing fitting and the frame member, and is flexible enough to deform to fit a shape of a gap between the part of the fixing fitting and the frame member.

[0008] According to the connector unit in the present disclosure, the external conductive component (for example, a round terminal and a bus bar) and the terminal fitting are fixed to the housing in a state of being fastened and fixed to the fixing fitting embedded in the housing and being electrically connected. Further, the heat transfer member is sandwiched between the part of the fixing fitting exposed from the housing and the metal frame member. Accordingly, the heat generated at the contact point between the terminal fitting and the counterpart terminal fitting, the connection location between the terminal fitting and the conductive component, and the like at the

time of energization is transferred in the order of the fixing fitting, the heat transfer member, and the frame member. The heat transfer member is flexible enough to deform to fit the shape of the gap between the frame member and the part of the fixing fitting, and thus comes into contact with both the frame member and the part of the fixing fitting with a larger contact area as compared with the case in which the heat transfer member does not have such flexibility. Further, even when an external force such as vibration is applied when the connector unit is used, or even when the connector unit is used for a long period of time, a state can be maintained in which the heat transfer member is in contact with the frame member and the part of the fixing fitting. In addition, the metal frame member has a large heat capacity and high thermal conductivity, and since the frame member is in contact with the outside air, the frame member also has excellent heat dissipation property. Accordingly, the connector unit having the present configuration can improve the heat dissipation property while avoiding an increase in size of the connector unit.

[0009] The heat transfer member may be in direct contact with the frame member and the part of the fixing fitting, or indirectly in contact with the frame member and the part of the fixing fitting with an adhesive, a pressure-sensitive adhesive, or the like interposed therebetween. However, in the latter case, it is preferable that the adhesive, the pressure-sensitive adhesive, or the like also has high thermal conductivity. Further, in addition to the adhesive, the pressure-sensitive adhesive, or the like, the heat transfer member may be indirectly in contact with the frame member and the part of the fixing fitting with another member that has high thermal conductivity or the like interposed therebetween.

[0010] The present disclosure has been briefly described above. Further, details of the present disclosure will be clarified by reading modes for carrying out the invention to be described below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawing which is given by way of illustration only, and thus is not limitative of the present disclosure and wherein:

[0012] FIG. 1 is a perspective view showing a connector unit and a counterpart connector according to an embodiment of the present disclosure;

[0013] FIG. 2 is a perspective view of the connector unit shown in FIG. 1;

[0014] FIG. 3 is a perspective view showing a state in which a housing, a frame member, and a heat transfer member constituting the connector unit shown in FIG. 2 are separated from one another;

[0015] FIG. 4 is a cross-sectional view taken along a line A-A in FIG. 2; and

[0016] FIG. 5 is an enlarged view of a portion B in FIG. 4.

DETAILED DESCRIPTION OF THE INVENTION

Embodiment

[0017] Hereinafter, a connector unit **1** according to an embodiment of the present disclosure will be described with reference to the drawings. The connector unit **1** shown in FIGS. 1 and 2 functions as a relay connector that electrically connects an electric wire **82** (see FIG. 4) extending from a housing **20** of the connector unit **1** and a counterpart connector **2** (see FIG. 1) fitted in the connector unit **1**.

[0018] Hereinafter, for convenience of description, “front”, “rear”, “upper”, “lower”, “left”, “right”, a “front-rear direction”, an “upper-lower direction”, and a “left-right direction” are defined as shown in FIG. 1. The “front-rear direction”, the “upper-lower direction”, and the “left-right direction” are orthogonal to one another. The front-rear direction coincides with the fitting direction

of the connector unit **1** and the counterpart connector **2**.

[0019] As shown in FIGS. **1** to **4**, the connector unit **1** mainly includes a terminal fitting **10**, the housing **20**, a nut member **30**, a frame member **40**, and a heat transfer member **50**. Hereinafter, the configurations of the members that constitute the connector unit **1** will be described in order.

[0020] First, the terminal fitting **10** will be described. As shown in FIG. **4**, the metal terminal fitting (the female terminal) **10** integrally includes a tubular contact portion **11**, a plate-shaped fastening portion **12** that is located on the rear side of the tubular contact portion **11**, and a plate-shaped connecting portion **13** that connects the tubular contact portion **11** and the plate-shaped fastening portion **12**. The tubular contact portion **11** is a portion that is electrically connected to a terminal fitting (a male terminal) **71** (see FIG. **1**) on the counterpart connector **2** side when the connector unit **1** and the counterpart connector **2** are fitted to each other, and has a tubular shape that extends in the front-rear direction. The plate-shaped fastening portion **12** is a portion to which an external terminal **81** connected to the electric wire **82** is fastened and fixed using the nut member **30** and a bolt **91**, and has a substantially rectangular flat plate shape whose plate thickness direction is oriented in the upper-lower direction (see also FIG. **1**). A bolt through hole **14** is formed in the central portion of the plate-shaped fastening portion **12**, penetrating the plate-shaped fastening portion **12** in the plate thickness direction (the upper-lower direction). A pair of left and right locking holes (penetrating holes) **15** are formed in the front end portion of the plate-shaped fastening portion **12** (see FIG. **1**).

[0021] Next, the housing **20** will be described. The housing **20** is a resin molded product, and as shown in FIGS. **3** and **4**, integrally includes an outer tube portion **21** that has a long hole shape elongated in the left-right direction as viewed from the front and that extends in the front-rear direction, a pair of cylindrical terminal fitting accommodation tube portions **22** that extend in the front-rear direction inside the outer tube portion **21** and that are arranged side by side in the left-right direction, and a flat plate-shaped rear end wall portion **23** that connects the rear end portions of the outer tube portion **21** and the pair of terminal fitting accommodation tube portions **22** to each other. The front end side of the outer tube portion **21** is open, and the rear end side of the outer tube portion **21** is blocked by the rear end wall portion **23**. The rear end side of each terminal fitting accommodation tube portion **22** is open, and the front end side of each terminal fitting accommodation tube portion **22** communicates with the outside through the through hole **24**. The tubular contact portion **11** of the terminal fitting **10** is accommodated in the terminal fitting accommodation tube portion **22**. A pair of upper and lower flat plate-shaped flange portions **25** that extend outwardly on both the upper and lower sides are provided at the front end edge portion of the outer tube portion **21**. A rubber packing **92** is provided on the outer peripheral surface of the outer tube portion **21** at a position adjacent to the rear side of the pair of flange portions **25**. When the housing **20** is mounted on the frame member **40**, the packing **92** has a function of sealing the gap between the inner peripheral surface of a penetrating hole **42** of the frame member **40** that is externally fitted onto the outer tube portion **21** and the outer peripheral surface of the outer tube portion **21**.

[0022] As shown in FIGS. **1**, **3**, and **4**, the housing **20** is integrally provided with a substantially rectangular flat plate-shaped extension portion **26** that extends rearward from a part of the rear end wall portion **23** adjacent to the lower side of the rear end openings of the pair of terminal fitting accommodation tube portions **22**. When the pair of left and right terminal fittings **10** are accommodated in the housing **20**, the pair of left and right plate-shaped fastening portions **12** cover the upper face of the extension portion **26** (see FIGS. **1** and **4**). When the pair of left and right terminal fittings **10** are accommodated in the housing **20**, as shown in FIGS. **4** and **5**, the nut member **30** is embedded (integrated) by insert molding in a location of the extension portion **26** where each of the pair of left and right bolt through holes **14** is formed. The nut member **30** is a member that functions to fasten and fix the terminal fitting **10** and the external terminal **81** to the housing **20**. The nut member **30** is made of metal, and includes a cylindrical tubular body portion

31 that extends in the upper-lower direction, that has a lower end side blocked by a bottom wall portion **31a**, and that has an open upper end side, and a flange portion **32** that extends from the opening edge portion at the upper end of the tubular body portion **31** in the radial direction of the tubular body portion **31**. A female screw corresponding to the male screw of the bolt **91** is formed on the inner peripheral side face of the tubular body portion **31**. A large portion of the tubular body portion **31** is embedded in the extension portion **26**, and the upper end opening of the tubular body portion **31** and the flange portion **32** are exposed to the outside at the upper face of the extension portion **26**. The bottom wall portion **31a** (more specifically, the lower end side portion of the bottom wall portion **31a**) of the tubular body portion **31** is exposed so as to slightly protrude downward from the lower end surface of the extension portion **26**.

[0023] A pair of left and right locking protrusions **27** are provided on the upper face of the extension portion **26** at positions adjacent to the front sides of the pair of left and right nut members **30**, corresponding to the pair of left and right locking holes **15** of the plate-shaped fastening portion **12** of each terminal fitting **10** (see FIG. 1). A plate-shaped partition wall portion **28** that protrudes upward and that extends in the front-rear direction is provided on the upper face of the extension portion **26** at a position between the pair of left and right nut members **30**. The partition wall portion **28** partitions the space between the pair of left and right plate-shaped fastening portions **12** to be provided on the upper face of the extension portion **26**, thereby functioning to prevent the occurrence of unintentional short circuits between the pair of left and right plate-shaped fastening portions **12**.

[0024] Next, the frame member **40** will be described. As shown in FIGS. 3 and 4, the metal frame member **40** includes a substantially rectangular flat plate-shaped body portion **41** that extends in the left-right direction and the upper-lower direction. In the central portion of the body portion **41**, the penetrating hole **42** that has a long hole shape elongated in the left-right direction as viewed in the front-rear direction and that penetrates the body portion **41** in the front-rear direction is provided corresponding to the outer tube portion **21** of the housing **20**. The outer tube portion **21** of the housing **20** is inserted into the penetrating hole **42**, so that the housing **20** is mounted on the frame member **40**.

[0025] As shown in FIGS. 1, 3, and 4, the frame member **40** is integrally provided with a substantially rectangular flat plate-shaped extension portion **43** that extends rearward from a part of the body portion **41** adjacent to the lower side of the rear end opening of the penetrating hole **42**. When the housing **20** is mounted on the frame member **40**, the extension portion **26** of the housing **20** covers the upper face of the extension portion **43** (see FIGS. 1 and 4). When the housing **20** is mounted on the frame member **40**, as shown in FIGS. 3 to 5, an installation surface **44** is provided at a location of the upper face of the extension portion **43**, which faces the bottom wall portions **31a** of the pair of left and right nut members **30** exposed from the lower end surface of the extension portion **26** with a small gap therebetween in the upper-lower direction. In this example, as shown in FIGS. 4 and 5, the installation surface **44** is a flat surface that extends so as to be slightly inclined with respect to the front-rear direction in an orientation in which the rear side is located higher than the front side, and that extends parallel to the left-right direction over the entire region of the extension portion **43** in the left-right direction. As will be described later, the installation surface **44** functions as a portion that sandwiches the sheet-shaped heat transfer member **50** with the bottom wall portion **31a** of the nut member **30**.

[0026] Next, the heat transfer member **50** will be described. The heat transfer member **50** is a member used to be sandwiched between the bottom wall portions **31a** of the pair of left and right nut members **30** exposed from the lower end surface of the extension portion **26** and the installation surface **44** of the frame member **40** when the housing **20** is mounted on the frame member **40** (see FIGS. 4 and 5). In this example, as shown in FIG. 3, the heat transfer member **50** has a thin sheet shape that is elongated in the left-right direction as viewed in the upper-lower direction. The heat transfer member **50** is made of a material that has higher thermal conductivity than the resin

material constituting the housing **20** and that is flexible enough to deform to fit the shape of the gap between the bottom wall portion **31a** of the nut member **30** and the installation surface **44** of the frame member **40**.

[0027] The heat transfer member **50** can be made of, for example, a thermally conductive resin material, a mixed material in which a resin serving as a base material is mixed with a heat transfer body having the thermal conductivity, and a mesh-shaped material made of a wire material having the thermal conductivity. The heat transfer member **50** may be formed by processing these materials into a plate shape or a tape shape. Further, if these materials are sufficiently soft under the usage environment of the heat transfer member **50**, these materials may be applied to the housing **20** or the frame member **40** in a paste form.

[0028] The configurations of the members that constitute the connector unit **1** have been described above.

[0029] Next, the assembling procedure of the connector unit **1** will be described. First, the pair of left and right terminal fittings **10** are accommodated in the housing **20**. Therefore, the tubular contact portion **11** of the terminal fitting **10** is inserted into each terminal fitting accommodation tube portion **22** of the housing **20** from the rear side. The plate-shaped fastening portion **12** of the terminal fitting **10** covers the upper face of the extension portion **26** of the housing **20** such that the bolt through hole **14** is located on the upper end opening of the nut member **30** and such that the locking hole **15** is locked to the locking protrusion **27** of the housing **20** (see FIGS. **1** and **4**). The locking hole **15** is locked to the locking protrusion **27**, so that the positional deviation of the terminal fitting **10** with respect to the housing **20** is prevented.

[0030] Next, the housing **20** is mounted on the frame member **40**. Therefore, the outer tube portion **21** of the housing **20** is inserted into the penetrating hole **42** of the frame member **40** until the flange portion **25** of the housing **20** abuts against the body portion **41** of the frame member **40** in a state in which the sheet-shaped heat transfer member **50** is attached to the bottom wall portions **31a** of the pair of left and right nut members **30** exposed from the lower end surface of the extension portion **26** of the housing **20** or the installation surface **44** of the extension portion **43** of the frame member **40**. When the housing **20** is completely mounted on the frame member **40**, the heat transfer member **50** is pressed and clamped between the bottom wall portion **31a** of the nut member **30** and the installation surface **44** of the frame member **40**, as shown in FIG. **5**, and due to the flexibility of the heat transfer member **50**, the heat transfer member **50** flexibly deforms to fit the shape of the gap between the bottom wall portion **31a** of the nut member **30** and the installation surface **44** of the frame member **40**. Therefore, as compared with a case in which the heat transfer member **50** does not have the flexibility, the heat transfer member **50** is in contact with both the nut member **30** and the frame member **40** with a large contact area. In other words, the heat transfer member **50** thermally connects the nut member **30** and the frame member **40**. Further, as described above, since the installation surface **44** extends so as to be slightly inclined with respect to the front-rear direction in an orientation in which the rear side is located higher than the front side, as compared with a case in which the installation surface **44** extends parallel to the front-rear direction, the heat transfer member **50** is less likely to be turned over due to the frictional force received from the bottom wall portion **31a** of the nut member **30** when the housing **20** is mounted on the frame member **40**, and the pressing force that the heat transfer member **50** receives from the bottom wall portion **31a** of the nut member **30** and the installation surface **44** increases, making it easier for the heat transfer member **50** to come into closer contact with the bottom wall portion **31a** of the nut member **30** and the installation surface **44**. According to the above, the assembly of the connector unit **1** is completed, and the connector unit **1** shown in FIGS. **1** and **2** is obtained.

[0031] The connector unit **1** after assembly is fitted into the counterpart connector **2** shown in FIG. **1**. The housing **60** of the counterpart connector **2** includes a connector portion **61** that extends in the front-rear direction and an electric wire accommodation portion **62** that extends in the upper-lower direction, and has a substantially L shape as viewed in the left-right direction. The connector

portion **61** includes an outer tube portion **63** having a shape that can be fitted into the connector portion **61** of the housing **20**, and a pair of left and right terminal fitting accommodation tube portions **64** each having a shape that can be fitted into a respective one of the pair of left and right terminal fitting accommodation tube portions **22** of the housing **20**. The terminal fitting (the male terminal) **71** is accommodated in each of the pair of left and right terminal fitting accommodation tube portions **64**. A pair of left and right electric wires **72**, one ends of which are connected to the pair of left and right terminal fittings **71**, pass through the inside of the electric wire accommodation portion **62** and extend downward to the outside from the lower end opening of the electric wire accommodation portion **62**.

[0032] The connector unit **1** and the counterpart connector **2** are fitted to each other such that the outer tube portion **21** is externally fitted onto the outer tube portion **63** and the terminal fitting accommodation tube portion **22** is internally inserted into the terminal fitting accommodation tube portion **64**. In a state in which fitting of the connector unit **1** and the counterpart connector **2** is completed, the tubular contact portion **11** of the terminal fitting **10** in the terminal fitting accommodation tube portion **22** is electrically connected to the terminal fitting **71** in the terminal fitting accommodation tube portion **64**. A packing **93** (see FIG. **1**) that is provided on the outer peripheral surface of the outer tube portion **63** has a function of sealing the gap between the inner peripheral surface of the outer tube portion **21** of the housing **20** that is externally fitted onto the outer tube portion **63** and the outer peripheral surface of the outer tube portion **63**.

[0033] Further, in the connector unit **1** after assembly, the external terminal **81** that is connected to the electric wire **82** is fastened and fixed to the plate-shaped fastening portions **12** of the pair of left and right terminal fittings **10** using the nut member **30** and the bolt **91** (see FIG. **4**). Specifically, the bolt **91** that is inserted into the bolt through hole **81a** formed in the flat plate-shaped portion of the external terminal **81** and the bolt through hole **14** of the plate-shaped fastening portion **12** in this order is screwed toward the internal space of the tubular body portion **31** of the nut member **30**, so that the terminal fitting **10** and the external terminal **81** are fastened and fixed to the nut member **30** fixed to the housing **20**.

[0034] In the connector unit **1**, the contact point between the terminal fitting **10** and the terminal fitting **71** is located in the housing **20** in order to be insulated from the outside even though the contact point is a location where the Joule heat generated in the terminal fitting is large at the time of energization due to the large contact resistance. Therefore, it is fairly difficult to dissipate heat from the contact point between the terminal fitting **10** and the terminal fitting **71** to the outside. Further, for example, when a large current passes through the connector unit **1** in the state in which the connector unit **1** and the counterpart connector **2** are fitted to each other, the amount of generated heat also increases. In this regard, in the connector unit **1**, the heat transfer member **50** is sandwiched between the bottom wall portion **31a** of the nut member **30** exposed from the housing **20** and the metal frame member **40**. Accordingly, the heat generated at the contact point between the terminal fitting **10** and the terminal fitting **71** at the time of energization is transferred in the order of the nut member **30**, the heat transfer member **50**, and the frame member **40**. Since the heat transfer member **50** has the flexibility as described above, the heat transfer member **50** comes into contact with both the frame member **40** and the nut member **30** with a large contact area as compared with a case in which the heat transfer member **50** does not have the flexibility. Accordingly, in the connector unit **1**, heat dissipation can be improved.

Operation and Effect

[0035] As described above, according to the connector unit **1** in the present embodiment, the external terminal **81** and the terminal fitting **10** are fixed to the housing **20** in a state of being fastened and fixed to the nut member **30** embedded in the housing **20** and being electrically connected. Further, the heat transfer member **50** is sandwiched between the bottom wall portion **31a** of the nut member **30** exposed from the housing **20** and the metal frame member **40**. Accordingly, the heat generated at the contact point between the terminal fitting **10** and the terminal

fitting **71** of the counterpart connector **2** at the time of energization is transferred in the order of the nut member **30**, the heat transfer member **50**, and the frame member **40**. The heat transfer member **50** is flexible enough to deform to fit the shape of the gap between the frame member **40** and the bottom wall portion **31a** of the nut member **30**, and thus comes into contact with both the frame member **40** and the bottom wall portion **31a** of the nut member **30** with a larger contact area as compared with the case in which the heat transfer member **50** does not have such flexibility. Further, even when an external force such as vibration is applied when the connector unit **1** is used, or even when the connector unit **1** is used for a long period of time, a state can be maintained in which the heat transfer member **50** is in contact with the frame member **40** and the bottom wall portion **31a** of the nut member **30**. In addition, the metal frame member **40** has a large heat capacity and high thermal conductivity, and since the frame member **40** is in contact with the outside air, the frame member **40** also has excellent heat dissipation property. Accordingly, the connector unit **1** according to the present embodiment can improve the heat dissipation property while avoiding an increase in size of the connector unit **1**.

Other Embodiments

[0036] The present disclosure is not limited to the embodiment described above and various modifications can be used within the scope of the present disclosure. For example, the present disclosure is not limited to the embodiment described above, and modifications, improvements, and the like can be appropriately made. In addition, the material, shape, size, number, arrangement position, and the like of the components in the embodiment described above are freely selected and are not limited as long as the present disclosure can be implemented.

[0037] Here, the features of the connector unit **1** according to the embodiment of the present disclosure described above are briefly summarized and listed in the following [1] to [3]. [0038] [1] A connector unit (**1**) including: [0039] a terminal fitting (**10**); [0040] a housing (**20**) configured to accommodate the terminal fitting (**10**); [0041] a fixing fitting (**30**) embedded in the housing (**20**) and configured to fasten and fix the terminal fitting (**10**) and an external conductive component (**81**); [0042] a metal frame member (**40**) on which the housing (**20**) is mounted; and [0043] a heat transfer member (**50**) configured to thermally connect the frame member (**40**) and a part (**31a**) of the fixing fitting (**30**) exposed from the housing (**20**), [0044] in which the heat transfer member (**50**) is sandwiched between the part (**31a**) of the fixing fitting (**30**) and the frame member (**40**), and is flexible enough to deform to fit a shape of a gap between the part (**31a**) of the fixing fitting (**30**) and the frame member (**40**).

[0045] According to the connector unit having the configuration in [1] described above, the external conductive component (for example, a round terminal and a bus bar) and the terminal fitting are fixed to the housing in a state of being fastened and fixed to the fixing fitting embedded in the housing and being electrically connected. Further, the heat transfer member is sandwiched between the part of the fixing fitting exposed from the housing and the metal frame member. Accordingly, the heat generated at the contact point between the terminal fitting and the counterpart terminal fitting, the connection location between the terminal fitting and the conductive component, and the like at the time of energization is transferred in the order of the fixing fitting, the heat transfer member, and the frame member. The heat transfer member is flexible enough to deform to fit the shape of the gap between the frame member and the part of the fixing fitting, and thus comes into contact with both the frame member and the part of the fixing fitting with a larger contact area as compared with the case in which the heat transfer member does not have such flexibility.

Further, even when an external force such as vibration is applied when the connector unit is used, or even when the connector unit is used for a long period of time, a state can be maintained in which the heat transfer member is in contact with the frame member and the part of the fixing fitting. In addition, the metal frame member has a large heat capacity and high thermal conductivity, and since the frame member is in contact with the outside air, the frame member also has excellent heat dissipation property. Accordingly, the connector unit having the present

configuration can improve the heat dissipation property while avoiding an increase in size of the connector unit.

[0046] The heat transfer member may be in direct contact with the frame member and the part of the fixing fitting, or indirectly in contact with the frame member and the part of the fixing fitting with an adhesive, a pressure-sensitive adhesive, or the like interposed therebetween. However, in the latter case, it is preferable that the adhesive, the pressure-sensitive adhesive, or the like also has high thermal conductivity. Further, in addition to the adhesive, the pressure-sensitive adhesive, or the like, the heat transfer member may be indirectly in contact with the frame member and the part of the fixing fitting with another member that has high thermal conductivity or the like interposed therebetween. [0047] [2] The connector unit (1) according to [1] described above, [0048] in which the frame member (40) has an installation surface (44) that extends in a direction intersecting a mounting direction in which the housing (20) is mounted on the frame member (40), and [0049] in which the heat transfer member (50) is sandwiched between the part (31a) of the fixing fitting (30) and the installation surface (44).

[0050] According to the connector unit having the configuration in [2] described above, when the housing is mounted on the frame member, the heat transfer member is sandwiched between the part of the fixing fitting and the installation surface of the frame member. Accordingly, the heat transfer member is sandwiched between the part of the fixing fitting and the installation surface and comes into close contact with the part of the fixing fitting and the installation surface. Here, since the installation surface extends in the direction intersecting the mounting direction in which the housing is mounted on the frame member, as compared with a case in which the installation surface extends parallel to the mounting direction, the heat transfer member is less likely to be turned over when the housing is mounted on the frame member, and the pressing force that the heat transfer member receives from the part of the fixing fitting and the installation surface increases, making it easier for the heat transfer member to come into closer contact with the part of the fixing fitting and the installation surface. Accordingly, the heat generated at the time of energization is efficiently transmitted from the part of the fixing fitting to the frame member through the heat transfer member. [0051] [3] The connector unit (1) according to [1] described above, [0052] in which a material constituting the heat transfer member (50) has higher thermal conductivity than a material constituting the housing (20).

[0053] According to the connector unit having the configuration in [3] described above, the material constituting the heat transfer member has higher thermal conductivity than the material constituting the housing. Accordingly, the heat generated at the connection location between the terminal fitting and the conductive component or the like can be dissipated to the outside through the heat transfer member more efficiently than when the housing is brought into direct contact with the shield member.

Claims

1. A connector unit comprising: a terminal fitting; a housing configured to accommodate the terminal fitting; a fixing fitting embedded in the housing and configured to fasten and fix the terminal fitting and an external conductive component; a metal frame member on which the housing is mounted; and a heat transfer member configured to thermally connect the frame member and a part of the fixing fitting exposed from the housing, wherein the heat transfer member is sandwiched between the part of the fixing fitting and the frame member, and is flexible enough to deform to fit a shape of a gap between the part of the fixing fitting and the frame member.
2. The connector unit according to claim 1, wherein the frame member has an installation surface that extends in a direction intersecting a mounting direction in which the housing is mounted on the frame member, and wherein the heat transfer member is sandwiched between the part of the fixing fitting and the installation surface.

3. The connector unit according to claim 1, wherein a material constituting the heat transfer member has higher thermal conductivity than a material constituting the housing.
